

## DAVID MURAGE

Software Developer | Information Technology Officer

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## SUMMARY

I am an IT Officer and Software Developer with expertise in both infrastructure management and application development. In my IT role, I maintain and optimize websites, manage cPanel, domains, DNS, hosting environments, and email systems, while ensuring strong server security and reliable digital operations.

On the development side, I build modern, responsive applications using the MERN stack (MongoDB, Express.js, React.js, Node.js), with experience in API integrations, Cloudinary for media management, and Dockerized deployments. This hybrid background allows me to bridge the gap between IT support and software engineering, delivering secure, scalable, and user-friendly digital solutions.

Passionate about technology, problem-solving, and innovation, I thrive in environments where I can strengthen IT systems while also creating impactful applications that drive organizational success.

Always learning. Always building. Always leveling up.

## EXPERIENCE

AVLC Group – Upper Hill, Nairobi, Kenya

### Information Technology Officer | Sep 2025 - Present

- Maintain and optimize company websites, providing back-end technical support to ensure smooth functionality and minimal downtime.
- Manage cPanel, domains, and email configurations, enabling efficient communication and seamless digital presence.
- Administer hosting environments and DNS settings, strengthening website/server security and overall system reliability.
- Support API integrations and connectivity, enhancing interoperability with third-party platforms and services.

## KALRO headquarters – Nairobi, Kenya

### ICT Volunteer | Feb 2025 – June 2025

- During my tenure as an ICT Volunteer at KALRO Headquarters, I provided essential technical and digital support to various departments, contributing to the organization's research and administrative efficiency.
- I was responsible for assisting staff with software installations and updates, where I diagnosed and resolved software-related issues while logging support tickets through a structured system. This ensured prompt service delivery, improved resolution tracking, and helped streamline internal IT support operations.
- I also maintained and regularly updated content on KALRO's official website and internal portals, uploading research publications, news items, staff announcements, and policy documents. My accuracy and attention to organizational tone and standards helped ensure that stakeholders had access to timely, consistent, and relevant information.
- A key part of my role involved supporting users of in-house applications and databases. I provided hands-on assistance to staff, helping them navigate system features, troubleshoot technical issues, and liaise with developers to address bugs. This improved user satisfaction and system adoption across departments.
- Additionally, I contributed to the integrity of KALRO's research and administrative data by accurately entering and validating large datasets into specialized portals. My approach supported high-quality data analysis and enhanced reporting capabilities, directly influencing informed decision-making in both scientific and operational domains.
- This volunteer experience significantly strengthened my technical problem-solving skills, my familiarity with content and database management systems, and my ability to communicate technical concepts to non-technical users. It also demonstrated my adaptability, attention to detail, and commitment to delivering practical IT solutions in a research-driven environment.

## KALRO Headquarters – Nairobi, Kenya

### ICT Attachee | Sep 2024 – Dec 2024

- **Provided technical support at the ICT help desk, assisting staff with hardware and software issues to ensure smooth operations:**  
At the ICT help desk, I delivered timely and effective technical support to staff, addressing a wide range of hardware and software issues. My responsibilities

included diagnosing computer malfunctions, resolving software compatibility problems, and ensuring proper configuration of devices. This proactive support minimized downtime, improved user productivity, and maintained a consistent level of operational efficiency across departments.

- **Gained hands-on experience in call center operations, managing inquiries and troubleshooting ICT-related challenges efficiently:**

I actively participated in call center operations where I handled a high volume of inquiries related to ICT issues. This experience sharpened my communication and problem-solving skills as I guided users through troubleshooting steps, escalated complex cases appropriately, and ensured every query was resolved with professionalism and urgency. My role contributed to a better end-user experience and strengthened the overall support structure.

- **Assisted in the maintenance and setup of computer systems, networks, and office technology equipment:**

I played a key role in the maintenance and setup of essential ICT infrastructure including desktop computers, printers, network routers, and other office technology tools. My tasks involved installing hardware components, configuring network connections, and ensuring devices were secure and operational. This hands-on involvement enhanced system reliability and ensured optimal performance of the workplace technology environment.

Zidio Development – Remote, Maharashtra India

### **Web Developer Intern | Feb 2024 – Mar 2024**

Worked as a backend developer and Implemented features like.

#### **1. User Authentication:**

As part of the backend development team, I was responsible for designing and implementing a secure user authentication system. This included allowing users to register, log in, and have their identities securely verified. I ensured that each account was uniquely associated with a user profile, enabling personalized access and secure data handling. This authentication system served as a foundational component for the application, supporting user-specific features and permissions.

#### **2. Real-Time Location Tracking:**

I developed a real-time location tracking feature that utilized the browser's Geolocation API to accurately capture and display the user's current position on the application interface. To enhance the user experience, I implemented a dynamic updating mechanism that continuously refreshed the user's location as they moved. This feature was particularly valuable for location-based services and was designed to operate efficiently across different devices and

environments.

### **3. Security and Privacy Measures:**

To uphold strong data security standards, I implemented robust authentication practices by integrating bcrypt.js for password hashing. This ensured that user credentials were securely stored and protected against unauthorized access. I followed best practices in backend development to safeguard sensitive information, helping the platform maintain high standards of user privacy and compliance.

## **EDUCATION**

Dedan Kimathi University of Technology – Nyeri, Kenya

Bachelor's in business information technology | Oct 2021 – May 2025

## **ACHIEVEMENTS & PROJECTS**

### **AI Accessibility Hackathon Participant – Impact Africa**

#### **Project: HuruSpaces – AI-Powered Accessibility Locator**

**Date: July 5, 2025**

Collaborated in a cross-functional team during the Impact Africa AI Hackathon to design and develop HuruSpaces, an innovative web platform aimed at supporting Persons With Disabilities (PWDs) in locating accessible venues such as restaurants, offices, hospitals, and shopping malls across urban centers.

Integrated a custom AI voice assistant named Huru AI, designed specifically to assist users with visual impairments and other disabilities. The assistant allows users to interact with the platform using voice commands, and provides real-time, personalized voice responses based on venue data and accessibility preferences.

Implemented intelligent filtering and categorization logic to recommend venues tailored to various disability needs (e.g., wheelchair accessibility, braille signage, guide dog-friendly areas, etc.).

Focused on inclusion, user-centered design, and AI-driven support, ensuring that the platform is usable, accessible, and helpful to marginalized populations in navigating everyday life.

The project showcased the use of AI for social impact, particularly in promoting digital inclusion, urban accessibility, and empowerment of underrepresented communities.

## **AI-Powered USSD Learning App — Hackathon Project, 2025**

### **Hackathon Winner – Impact Africa x Groq (2025)**

- Co-developed an AI-powered educational quiz system that delivered real-time math questions aligned with the Kenyan primary school curriculum, targeting students in low-connectivity rural areas.
- Implemented an intelligent point-based scoring system that rewards students for correct answers, promoting engagement and continuous learning.
- Students interacted with the system through USSD (Unstructured Supplementary Service Data), allowing participation via basic feature phones without internet access.
- Leveraged Groq API and LLM prompt engineering to dynamically generate and adapt quiz content based on student input and curriculum standards.
- Built and deployed the backend using Flask and Python, handling session flows, question logic, and scoring in real-time.
- Won 1st place for innovation and social impact at the Impact Africa x Groq Hackathon, recognized for demonstrating how AI can be applied in low-resource environments to advance digital inclusion in education.

Tools & Technologies: Python, Flask, Groq API, USSD, LLM Prompt Engineering

## **SKILLS**

### **Technical Skills:**

- **Programming Languages:**  
Proficient in JavaScript, with the ability to write clean, efficient, and scalable code for both frontend and backend development.
- **Frameworks & Libraries:**  
Experienced in using React.js to build dynamic and responsive user interfaces, and Node.js for building fast and scalable server-side applications.

- **Tools & Technologies:**

Skilled in working with MongoDB for NoSQL database management and Git for version control and collaborative development workflows.

- **Additional Skills:**

Strong understanding of RESTful API integration, responsive web design principles, and third-party payment gateway integration including Daraja API (M-Pesa) and Stripe. Familiar with end-to-end application development and deploying full-stack web solutions.

**Soft Skills:**

- Problem-solving, teamwork
- Team Collaboration & Facilitation

## References

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